

InfraPulse — Automated IT Compliance & Asset Inspection Portal

Engineered by Prevoyance IT Solutions

A full-stack, enterprise-grade IT compliance auditing and asset management platform. Discovers, inspects, and generates structured compliance reports for Windows, macOS, and Linux workstations across local networks and remote internet locations with zero manual configuration.

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Project Name

InfraPulse: Enterprise IT Compliance Audit & Asset Inspection System

Engineered by: **Prevoyance IT Solutions**

Overview

The **InfraPulse IT Compliance Portal** is a centralized, self-hosted audit solution engineered by **Prevoyance IT Solutions**. It automates system inventory collection, hardware verification, software auditing, security policy inspection, and compliance report generation across all workstation operating systems (Windows, macOS, Linux).

Whether laptops are connected on the same local Wi-Fi router or operating remotely from home offices across different cities, InfraPulse allows IT administrators and auditors to gather instant, tamper-proof hardware and software inspection reports.

Problem Statement

Traditional IT compliance audits at branch offices and corporate workstations face major operational hurdles:

1. **Time-Consuming Manual Audits:** IT engineers must physically log into every employee laptop, run commands, check control panel software, and write manual Excel reports.
2. **Remote & Work-From-Home Invisibility:** Laptops operating outside the local office network or on different Wi-Fi routers cannot be scanned using standard local IP range tools.
3. **Complex Terminal Commands for Non-Technical Staff:** Employees are reluctant or unable to open PowerShell or Terminal to run raw CLI commands.
4. **Data Security & Link Exposure:** Standard audit scripts expose backend IP addresses and sensitive API endpoints in plain text inside terminals or browsers.
5. **Lack of Automated Change Tracking:** Difficulty in identifying when an employee installs unauthorized software or replaces a RAM module or hard drive between quarterly audits.

InfraPulse solves every single one of these problems with 1-click double-clickable launchers, Base64 stealth obfuscation, multi-OS support, real-time cloud tunnels, and automated PDF compliance reports.

Key Features

- **Cross-Platform Support:** Audits Windows 10/11, macOS (Intel & Apple Silicon), and Linux (Ubuntu, Debian, RHEL, WSL).
- **1-Click Native Executable Launchers:** Downloadable double-clickable files (`.exe` for Windows, `.command` for macOS Finder, `.sh` for Linux Desktop) so users never have to touch a terminal.
- **Stealth Base64 Command Obfuscation:** Terminal deployment one-liners hide all URLs and API endpoints. Direct browser access is automatically blocked (`404 Not Found`).
- **Automated PDF & XML Compliance Reports:** Generates professional, branded PDF inspection reports containing hardware specs, software inventory, hotfixes, antivirus status, and login history.
- **Smart Network Scanner Engine:** Parallel ICMP Ping Sweep (128 threads), TCP Port Scanning (64 threads), and ARP discovery for subnet device identification.
- **Software & Hardware Delta Diff Engine:** Automatically compares consecutive scans of the same laptop to highlight newly installed software, removed apps, and hardware modifications.
- **WiFi Dashboard & Metrics:** Real-time signal strength analysis and indoor router distance estimation using RSSI log-distance path loss modeling.

Tech Stack

Layer	Technologies Used	Purpose
Backend Core	Python 3.10+ / FastAPI / Uvicorn	High-performance async REST API & ingestion server
Frontend UI	HTML5, Vanilla CSS3 (Custom Glassmorphism), ES6 JS	Rich single-page dashboard with zero heavy external dependencies
Database	SQLite3 + WAL Mode	Lightweight, real-time relational storage for device audits
Report Generator	ReportLab PDF Engine	Automated PDF audit report generation
Windows Executable Compiler	Microsoft C# Compiler (<code>csc.exe</code>) / PowerShell VBS	Dynamic <code>.exe</code> and <code>.vbs</code> binary launcher generator

Layer	Technologies Used	Purpose
Unix Engine	POSIX Shell (bash) / Python3 Fallback	Agentless macOS and Linux data collection script
Network & Security	socket , subprocess , Base64, WinHTTP CLI	Subnet scanning, ARP parsing, and obfuscated communication

Portal Workflow Guide

1. Info Gathering (Network Scanner)

The **Info Gathering** tab allows IT officers to discover all active devices on the local subnet without installing agents on target machines.

- **Subnet Scanning:** Automatically populates the local network range (e.g., 192.168.1.0/24).
- **Multi-Phase Scan:** Runs parallel ICMP ping sweeps and TCP port scans (ports 22, 80, 135, 445, 3389, 9100).
- **Device Classification:** Automatically classifies detected devices into Windows Workstations, Linux Servers, Web Services, Printers, and Mobile Devices.
- **Audit Status Badges:** Displays whether each discovered device is already audited or unaudited.

2. Device Audits

The **Device Audits** tab provides deep hardware, software, and compliance inspection for any selected workstation.

- **Device Selection:** Choose any audited device from the dropdown menu.
- **System Overview:** Inspect CPU, RAM, storage partitions, motherboard model, BIOS version, serial number, and operating system build.
- **Software Inventory:** View, search, and filter every installed application along with publisher name, version, installation date, and storage size.
- **Login History:** Review recent interactive logons collected from the Windows Security Event Log.
- **Hardware & Software Diff:** Automatically compares the latest audit against previous scans to highlight newly installed applications, removed software, or replaced hardware components.

- **PDF Inspection Report:** Click **Download PDF Report** to generate a branded compliance report for documentation.

3. WiFi Dashboard

The **WiFi Dashboard** manages network connectivity and proximity metrics.

- **Available WiFi Networks:** Scans nearby wireless networks, displaying SSID names, signal strength percentages, and encryption types.
- **Current Connection Metrics:** Shows active connection status, local IP address, subnet mask, and gateway details.
- **Distance Estimation:** Calculates estimated physical distance (in meters) between the workstation and the Wi-Fi access point using log-distance RSSI path loss modeling.

4. All Devices

The **All Devices** tab serves as the central inventory registry for the organization.

- **Master Device List:** Displays a complete, filterable table of every machine that has completed a compliance audit.
- **Quick Overview:** View device name, operating system type, last logged-in user, and exact timestamp of the last audit.
- **Direct Audit Link:** Click **View Audit** on any row to open the complete inspection view for that device.

5. Terminal Command

The **Terminal Command** tab handles deployment configuration and executable launcher downloads.

- **Deployment Mode Configuration:** Set custom server IP addresses and ports for local network or remote cloud deployment.
- **Obfuscation Toggle:** Enables Base64 encoding for one-line terminal execution commands to hide backend URLs.
- **Double-Clickable Launchers:** Provides direct 1-click download links for compiled binary launchers (`.exe` for Windows, `.command` for macOS, `.sh` for Linux).

6. Settings

The **Settings** tab manages portal configuration and administrative tools.

- **Portal Preferences:** Configure default audit parameters, report branding, and storage options.
- **Asset Metadata Management:** Edit or assign custom asset tags, owner details, department locations, and lifecycle stages (Active, Maintenance, Retired).

Installation & Execution Guide (Step-by-Step for All 3 OS)

1. Windows Setup & Execution (.exe / .vbs)

1. On the target Windows laptop, open the portal and navigate to the **Terminal Command** tab.
2. Click **Windows Binary (.exe Launcher)** to download `RunAudit_Windows.exe` (or click **Windows VBS (.vbs)** to download `RunAudit_Windows.vbs`).
3. Open the **Downloads** folder and **double-click** `RunAudit_Windows.exe` .
4. The audit will run silently in the background with zero terminal windows popping up. The report will appear on the dashboard within 10 seconds.

2. macOS Setup & Execution (.command)

1. Open the portal dashboard and navigate to the **Terminal Command** tab.
2. Click **macOS Launcher (.command)** to download `RunAudit_Mac.command` .
3. Open Terminal in your Downloads folder and grant execution permission:

```
chmod +x RunAudit_Mac.command
```

4. Execute the launcher:

```
./RunAudit_Mac.command
```

5. *(Optional Finder Execution)*: Alternatively, you can double-click `RunAudit_Mac.command` directly in Finder. If macOS displays a security prompt, right-click the file, select **Open**, and click **Open**.

6. The script executes the audit scan and submits the report automatically.

3. Linux Setup & Execution (.sh)

1. Open the portal dashboard and navigate to the **Terminal Command** tab.
2. Click **Linux Launcher (.sh)** to download `RunAudit_Linux.sh`.
3. Open the terminal in your Downloads folder and grant execution permission:

```
chmod +x RunAudit_Linux.sh
```

4. Execute the launcher:

```
./RunAudit_Linux.sh
```

5. The audit data is collected and uploaded directly to your compliance portal.

Important Testing Note

Note: During testing, **Prevoyance IT Solutions** will provide a live, pre-configured **Cloudflare Tunnel URL** (e.g., <https://xxxx.trycloudflare.com>).

Simply enter this provided Cloudflare link into Section 3 of your dashboard (**Terminal Command** tab), and all generated commands and double-clickable launchers will work over HTTPS across any laptop worldwide without requiring router port forwarding or manual IP setup.

Thank You for Choosing Prevoyance IT Solutions

Delivering Smart, Automated & Secure IT Infrastructure Solutions.

*For technical assistance, custom deployment inquiries, or security compliance support, please reach out to the **Prevoyance IT Solutions** technical team.*